



Wiz's Journey: Mastering the States of Matter

Producing a Serious Game to Aid in Teaching Matter and Their Phase Changes to Grade 6 Learners

A Thesis

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ABSTRACT

Chemistry education contains foundational concepts that are often perceived as difficult for students to understand, particularly in terms of their application to real-world situations. This difficulty may lead to cognitive overload, especially when instruction lacks opportunities for practical engagement. Interactive learning approaches allow students to actively apply and develop their analytical skills while discovering how chemistry concepts relate to everyday experiences.

This study explores the use of Game-Based Learning (GBL) through the development of *Wiz's Journey: Mastering the States of Matter*, an educational serious game designed as a learning material for Grade 6 students under the MATATAG curriculum in the Philippines. The game focuses on physical and chemical changes of matter, expanding students' understanding beyond the three common states of matter. Through interactive gameplay, students engage with concepts such as phase changes and heat transfer while learning to distinguish physical changes from chemical changes.

The primary goal of this study is to improve students' conceptual understanding, motivation, and emotional engagement by reinforcing hands-on learning and real-world applications of chemistry concepts. This project aims to provide both students and teachers with an accessible and engaging instructional resource that supports visualization, experimentation, and the development of analytical skills in basic chemistry education.

Keywords: physical and chemical changes, game-based learning, educational serious game

Sustainable Development Goals: 4: Quality Education, 11: Sustainable Cities and Communities, 12: Responsible Consumption and Production

ABSTRAK

Ang pag-aaral ng kimika ay naglalaman ng mga pundamental na konsepto na kadalasang itinuturing na mahirap unawain ng mga mag-aaral, lalo na pagdating sa aplikasyon nito sa totoong buhay. Ang ganitong kahirapan ay maaaring magdulot ng cognitive overload, partikular kung kulang ang mga aralin sa praktikal at interaktibong gawain. Sa pamamagitan ng interaktibong pagkatuto, aktibong nahahasa ng mga mag-aaral ang kanilang kasanayan sa pagsusuri at mas nauunawaan nila kung paano nagagamit ang mga konsepto ng kimika sa pang-araw-araw na buhay.

Sinisiyasat ng pag-aaral na ito ang paggamit ng Game-Based Learning (GBL) sa pamamagitan ng pagbuo ng *Wiz's Journey: Mastering the States of Matter*, isang pang-edukasyong seryosong laro na idinisenyo bilang pantulong na materyal sa pagkatuto para sa mga mag-aaral sa Baitang 6 sa ilalim ng MATATAG kurikulum sa Pilipinas. Nakatuon ang laro sa pisikal at kemikal na pagbabago ng materyal, na nagpapalawak sa kaalaman ng mga mag-aaral lampas sa tatlong karaniwang kaanyuan ng matter. Sa pamamagitan ng interaktibong gameplay, natutuhan ng mga mag-aaral ang mga konsepto tulad ng pagbabago ng estado at paglipat ng init, habang malinaw na natutukoy ang kaibahan ng pisikal at kemikal na pagbabago.

Layunin ng pag-aaral na ito na mapabuti ang pag-unawa, motibasyon, at emosyonal na pakikipag-ugnayan ng mga mag-aaral sa pamamagitan ng hands-on na pagkatuto at aplikasyon ng mga konsepto ng kimika sa mga sitwasyong hango sa totoong buhay. Nilalayon din ng proyektong ito na magsilbing kapaki-pakinabang na materyal para sa mga guro at mag-aaral upang mas mapadali ang pagkatuto ng kimika sa isang biswal, interaktibo, at makabuluhang paraan.

Mga susing salita: pisikal at kemikal na pagbabago, game-based learning, pang-edukasyon na laro

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Science education at the elementary level serves as a foundational stage in developing scientific literacy, conceptual reasoning, and problem-solving skills. Within the MATATAG Curriculum of the Department of Education (DepEd), Grade 6 Science introduces learners to fundamental concepts related to materials, including the description of changes of state for solids, liquids, and gases—specifically melting, evaporation, freezing, and condensation—using diagrams and flowcharts. These competencies aim to help learners understand observable phenomena through scientific representations and structured reasoning.

Despite the importance of these competencies, research has consistently shown that students experience difficulty in understanding the particulate nature of matter and the mechanisms underlying phase changes. Learners often memorize definitions of melting or evaporation without fully comprehending the particle-level interactions that explain these transformations. Misconceptions frequently arise regarding particle spacing, motion, and the role of heat transfer in altering states of matter (Adadan, 2013; Paik, Kim, Cho, & Park, 2004). These challenges are compounded by the abstract nature of microscopic representations, which are not directly observable.

In many classroom settings, demonstrations of phase change are limited to water-based experiments, such as melting ice or boiling water. While these activities provide observable macroscopic changes, they often lack explicit integration with particle-level models, resulting in fragmented conceptual understanding. Furthermore, access to laboratory materials for demonstrating processes such as sublimation or deposition is limited at the elementary level, restricting exposure to diverse examples of state change.

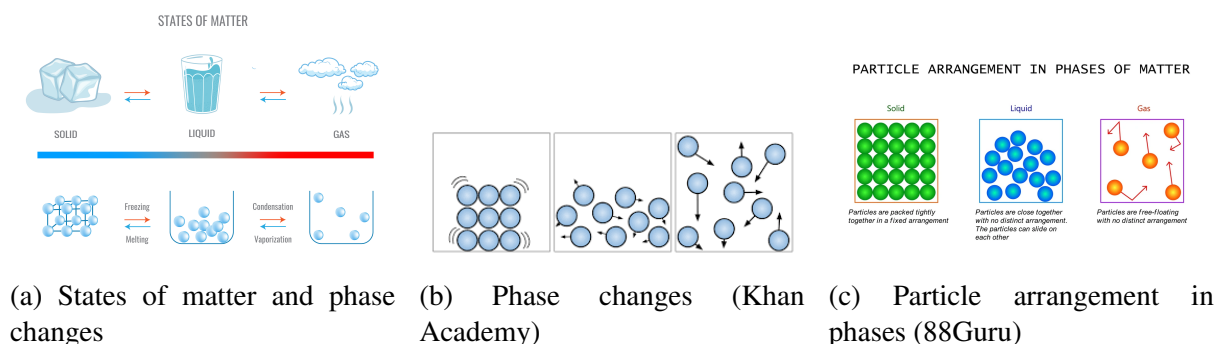


Figure 1.1: Examples of phase change diagrams used in science education

Game-Based Learning (GBL) has emerged as a promising instructional approach that integrates interactive systems with curricular objectives. Educational games can provide dynamic visualizations, immediate feedback, and experiential learning opportunities that reduce cognitive overload and promote conceptual change (Plass, Homer, & Kinzer, 2015). When designed with explicit learning outcomes and aligned assessments, digital games may enhance engagement and deepen understanding, particularly in abstract scientific domains.

This study introduces *Wiz’s Journey: Mastering the States of Matter*, a 2D educational platformer developed as a supplementary and enrichment learning tool aligned with the MATATAG Grade 6 Science competency on changes of state. The game integrates explicit terminology (e.g., “Solid → Liquid: Melting”) while allowing learners to discover appropriate phase changes through environmental problem-solving. Central to its design is an interactive particle-lattice minigame—a core mechanic of the gameplay—in which learners manipulate bonds and particle spacing to simulate state transitions. By directly engaging with representations similar to those used in classroom diagrams, learners repeatedly encounter visual models of particle behavior in a contextualized, goal-driven environment.

Given the increasing emphasis on learner-centered instruction under the MATATAG framework, there is a need to explore innovative digital tools that complement formal instruction without replacing it. This study examines whether a platformer-based educational game can support conceptual understanding, engagement, and reinforcement of curriculum-aligned competencies related to phase changes of matter.

1.2 Objectives

1.2.1 General Objective

This study aims to design, develop, and evaluate an educational platformer game that rapidly operationalizes a selected learning competency from the first quarter of the Grade 6 MATATAG Science curriculum. The game will be developed with established pedagogical and assessment strategies in mind to ensure alignment with curriculum standards and learning outcomes. Learner performance and engagement will be evaluated through pretest and posttest analyses to determine the effectiveness of the intervention.

1.2.2 Specific Objectives

Specifically, the study aims to:

1. Design learning outcome mechanics (LOMs) that strategically target both lower-order and higher-order cognitive learning outcomes aligned with the MATATAG Grade 6 Science competency on changes of state.
2. Develop a functional 2D educational platformer game that incorporates the designed LOMs and integrates interactive particle-level simulations to visually represent state transitions.
3. Evaluate the game's efficacy in teaching the selected learning outcomes to Grade 6 learners through structured pre- and post-assessment instruments, engagement surveys, and qualitative feedback from students regarding usability and perceived learning value.

1.3 Scope and Limitations of the Study

This study focuses exclusively on the development and evaluation of *Wiz's Journey: Mastering the States of Matter*, a 2D educational platformer designed to reinforce the Grade 6 Science competency from the MATATAG Curriculum Guide (CG) 2023 for Grades 3-10. Specifically, the game targets the Quarter 1: Materials learning competency that requires students to “describe changes of state for solids, liquids, and gases as melting, evaporation, freezing, and condensation using diagrams and flowcharts.” The game addresses physical changes only and does not include instruction on chemical reactions or other chemical changes.

The game mechanics are designed to reflect the chosen learning outcomes, focusing on interactive methods for handling game content that directly map to the MATATAG competency. During the development phase, which commenced in December 2025 and extended through March 2026, the development team reviewed existing educational materials and pedagogical approaches to identify optimal mechanics and instructional strategies that could be contextualized for Grade 6 students' current knowledge of phase changes.

The game was developed using Unity as the game engine, chosen for its cross-platform compatibility and the development team's proficiency with the tool. The game is deployed on multiple platforms for greater accessibility: primarily on Android mobile devices via Itch.io, as well as on web platforms via Unity Play. This multi-platform approach ensures that the game can reach students regardless of their available device.

The evaluation methodology consisted of two phases: a formative playtest involving a small sample of students in a controlled environment with qualitative observations and surveys for iterative improvements, followed by a summative playtest with a larger sample that incorporated pretest, gameplay, and posttest assessments to measure learning outcomes.

However, recruitment and participation differed from the initially planned scope. The study originally planned to recruit participants from nearby schools including Inchican Elementary School, Seven Pillars Catholic School, and Princeton Academy. In actuality, only Inchican Elementary School was able to provide informed consent and participate in the study. The school provided 5 consenting students: 3 participants for the formative playtest and 18 participants for the summative playtest (compared to the initially promised 30 students). The other targeted schools were unable to participate due to logistical constraints, including lack of response to communication attempts and the timing of the study during the end of the academic school year when students were occupied with moving-up ceremonies from Grade 6 to first-year high school.

The study does not measure long-term retention of learning outcomes or academic performance beyond the duration of the intervention. Additionally, the research evaluates the game as a supplementary enrichment tool and does not employ a control group design, limiting the ability to make direct comparisons with traditional classroom instruction. The study adopts a single-group quasi-experimental design with pre- and post-assessments, engagement surveys, and qualitative interviews focused on student perception and usability feedback.

1.4 Significance of the Study

This study contributes to elementary science education in the Philippines by examining the integration of interactive digital modeling within a platformer game environment. For learners, *Wiz's Journey: Mastering the States of Matter* provides repeated exposure to particle-level representations within meaningful problem-solving contexts, strengthening conceptual connections between diagrams and observable phenomena related to phase changes. By aligning the game with the MATATAG Grade 6 Science competency, the study demonstrates how educational games can reinforce curriculum-aligned learning objectives in accessible, engaging ways.

For educators and school administrators, this study offers insights into how educational games may complement MATATAG-aligned instruction without replacing formal teaching. The findings may inform future curriculum integration strategies, digital resource development, and professional development initiatives for teachers adopting game-based learning approaches. Given the resource constraints faced by many Philippine schools, this game's availability on multiple platforms (web and mobile) addresses accessibility barriers and demonstrates feasibility of deploying educational technology in resource-limited contexts.

For researchers and the broader educational community, this study adds to the growing body of literature on game-based learning in science education, particularly within the

Philippine educational context and addressing the K-12 curriculum transition. By examining a discovery-based yet explicitly labeled game design combined with systematic evaluation using pre-post assessments and engagement surveys, the study contributes to ongoing discussions on balancing constructivist learning principles with guided instruction in digital environments. The qualitative feedback from student participants and actual implementation challenges encountered (such as participant recruitment constraints) provide realistic insights into deploying educational games in authentic school settings.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Review of Related Literature

2.1.1 Conceptual Change in Science Education

Learners frequently enter science classrooms with pre-existing misconceptions about natural phenomena. In the context of phase changes, students often misunderstand particle behavior, energy transfer, and state transitions at the molecular level. Conceptual Change Theory posits that meaningful learning occurs when learners confront inconsistencies in their existing mental models and reconstruct their understanding based on intelligible and plausible alternatives.

Interactive simulations have been shown to support conceptual restructuring by making abstract microscopic processes visible. By allowing learners to observe and manipulate particle arrangements during phase transitions, digital environments can facilitate deeper conceptual understanding and correction of misconceptions.

This theoretical perspective informs the educational foundation of the present study.

2.1.2 Constructivism and Interactive Learning

Constructivist Learning Theory asserts that knowledge is actively constructed through interaction and experience rather than passively received. Game-based environments embody constructivist principles by placing learners in active problem-solving roles.

In platform-based game environments, players experiment, test strategies, and receive immediate feedback from system responses. This aligns with the constructivist view that learning occurs through engagement and reflection upon action.

The interactive particle manipulation mechanics of the developed game reflect this constructivist orientation.

2.1.3 Multimedia Learning in Science Education

Multimedia Learning Theory suggests that learning is enhanced when verbal and visual information are presented in coordinated and complementary ways. Effective multimedia design integrates text, visuals, and animation to support meaningful cognitive processing.

The game integrates explicit textual labeling (e.g., “Solid → Liquid: Melting”) alongside dynamic molecular visualization. This dual-channel presentation supports the formation of coherent mental representations of microscopic processes.

2.1.4 Game-Based Learning and Engagement

Game-Based Learning Theory emphasizes the role of challenge, feedback, and goal-oriented interaction in promoting learner engagement. Educational games combine instructional objectives with gameplay mechanics to foster sustained attention and intrinsic motivation.

Engagement is a critical factor in educational technology adoption. Games that balance challenge and usability are more likely to maintain player interest and promote exploratory interaction.

This study evaluates engagement using structured post-play instruments to determine whether the finalized build demonstrates positive user experience.

2.1.5 Experiential Learning and Simulation

Experiential Learning Theory posits that learning occurs through cycles of experience, reflection, conceptualization, and experimentation. Simulated environments allow learners to experience phenomena that may not be directly observable in traditional classrooms.

By simulating molecular interactions during freezing, melting, and evaporation, the game provides experiential exposure to abstract scientific processes.

2.1.6 Outcome-Based Education, Bloom’s Taxonomy, and MATATAG Curriculum

Outcome-Based Teaching and Learning emphasizes alignment between instructional activities and clearly defined competencies. The MATATAG Curriculum for Grade 6 Science outlines competencies related to understanding states of matter and phase transitions.

Bloom’s Taxonomy further categorizes cognitive processes from remembering to applying and analyzing. The game’s level design encourages players to apply conceptual knowledge to solve in-game environmental challenges, aligning with higher-order cognitive objectives.

2.1.7 Iterative Design and User-Centered Game Development

In game development research, iterative design is a widely accepted approach for refining gameplay mechanics, usability, and user experience. Iterative development involves repeated cycles of testing, evaluation, modification, and regression validation.

Campaign testing refers to structured playthrough sessions aimed at identifying usability issues, progression bottlenecks, and gameplay friction points. Regression testing ensures that modifications introduced in later builds do not reintroduce previously resolved issues.

User-centered design principles emphasize incorporating player feedback into successive iterations to improve overall game quality.

The present study adopts an iterative development framework prior to conducting a final structured evaluation of the game's usability and engagement.

2.2 Review of Related Studies

2.2.1 International Studies

International research has demonstrated the effectiveness of educational games in enhancing engagement and conceptual understanding in science subjects. Studies evaluating interactive simulations and digital learning games report improvements in learner motivation and positive user experience when multimedia elements and active manipulation are integrated.

Additionally, research on usability evaluation models such as MEEGA+KIDS highlights the importance of assessing engagement, focused attention, satisfaction, and perceived learning in educational game contexts.

Studies on iterative game development emphasize that structured playtesting contributes significantly to improved playability, stability, and user satisfaction.

2.2.2 Local Studies

Local studies in educational technology within the Philippine context indicate increasing integration of digital learning tools aligned with curriculum standards. Research evaluating instructional games commonly employs usability surveys and engagement instruments to assess learner reception.

However, limited research combines structured iterative development practices with formal engagement evaluation within educational game development projects.

This gap supports the relevance of the present study.

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2.4 Theoretical Framework

This study is grounded in Conceptual Change Theory, Constructivism, and Multimedia Learning Theory as primary anchors explaining how interactive simulation may support scientific understanding.

Supporting theories include Game-Based Learning Theory and Experiential Learning Theory, which justify the use of challenge-based interactive mechanics to foster engagement.

Outcome-Based Teaching and Learning principles and Bloom’s Taxonomy guide the alignment of game mechanics with Grade 6 MATATAG competencies.

In addition, iterative design principles and user-centered development frameworks support the methodological choice of conducting campaign and regression testing prior to formal usability evaluation.

Together, these frameworks explain both:

1. The pedagogical rationale of the game, and
2. The developmental methodology employed in refining the intervention.

2.5 Conceptual Framework

The study adopts an expanded Input–Process–Output (IPO) model integrating both pedagogical foundations and iterative development practices.

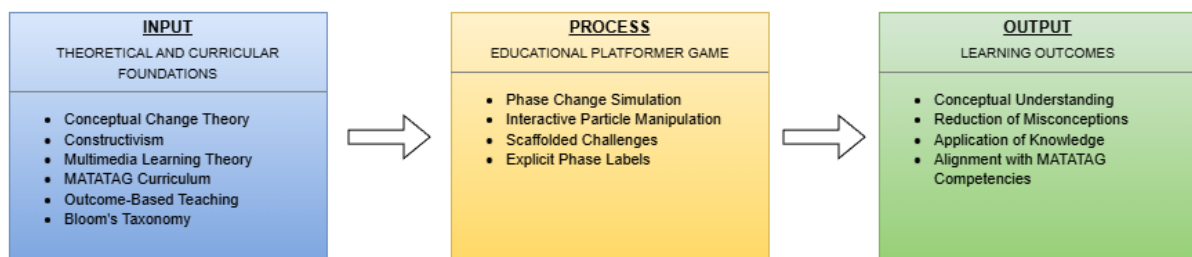


Figure 2.1: Conceptual Framework of the Study

2.5.1 Explanation of the Framework

The **Input** component consists of theoretical foundations (Conceptual Change, Constructivism, Multimedia Learning, Game-Based Learning, Experiential Learning), curricular alignment (MATATAG Curriculum and Bloom’s Taxonomy), and user-centered design principles.

The **Process** component is divided into two phases:

1. Iterative Development Phase

- Campaign testing

- Regression testing
- Playtesting feedback integration
- Gameplay refinement

2. **Final Evaluation Phase**

- Structured play session using finalized build
- MEEGA+KIDS usability and engagement assessment
- Qualitative feedback collection

The **Output** component includes:

- Improved gameplay stability
- Refined user experience
- Measured engagement levels
- Evaluated usability of the finalized educational platformer

The framework illustrates that pedagogical theory informs game design, iterative development refines implementation, and structured evaluation measures user engagement and usability.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Research Design

This study employed a developmental experimental research design integrating iterative game development practices with structured usability and engagement evaluation. The research was conducted in two major phases: (1) Iterative Development and Testing, and (2) Final Structured Evaluation.

The first phase focused on refining the educational platformer through multiple campaign and regression testing cycles. The second phase involved evaluating the finalized build using standardized usability and engagement instruments.

This design was selected to align with user-centered game development methodologies while maintaining academic rigor in assessing the educational game's usability and engagement potential.

3.2 Development Framework

The development of the educational platformer followed an iterative design approach grounded in user-centered development principles.

3.2.1 Iterative Development Cycles

The game underwent multiple structured development cycles consisting of:

1. **Build Implementation** – Addition or refinement of gameplay mechanics.
2. **Campaign Testing** – Structured playthrough sessions to identify usability issues, progression bottlenecks, gameplay friction, and feature clarity.
3. **Regression Testing** – Verification that newly implemented changes did not reintroduce previously resolved issues.
4. **Feedback Integration** – Refinement of mechanics, level design, UI clarity, and performance stability.

Each iteration was documented using testing logs detailing:

- Identified bugs
- Gameplay balancing concerns
- Player confusion points
- Feature stability issues
- Suggested improvements

Data from this phase were analyzed descriptively to track improvement trends across builds.

3.3 Participants

Participants were selected using convenience sampling. Approximately 20–30 individuals were recruited through personal networks, academic peers, and referrals.

Participants were not restricted to Grade 6 students due to timeline and accessibility constraints. However, respondents were considered representative users capable of evaluating usability, engagement, and clarity of the educational mechanics.

Where participants were minors, parental consent was secured prior to participation.

Different participants may have participated across iterative testing cycles. However, the final structured evaluation phase was conducted using a stable release candidate build.

3.4 Research Instruments

The study utilized both developmental testing tools and structured evaluation instruments.

3.4.1 Campaign and Regression Testing Logs

Structured testing logs were used to document:

- Bug occurrence frequency

- Severity classification (minor, moderate, critical)
- Gameplay progression issues
- UI clarity concerns
- Player navigation difficulties
- Performance stability

These logs enabled systematic tracking of improvements across development cycles.

3.4.2 MEEGA+KIDS Usability and Engagement Instrument

The finalized version of the game was evaluated using the MEEGA+KIDS model. This instrument measures:

- Fun
- Focused attention
- Perceived learning
- Usability
- Challenge
- Satisfaction

Participants responded using a Likert-scale format.

3.4.3 Qualitative Feedback Questionnaire

Participants provided open-ended responses regarding:

- Strengths of the game
- Areas requiring improvement
- Suggestions for refinement
- Perceived clarity of phase change concepts

3.5 Data Collection Procedure

The research was conducted in two phases:

3.5.1 Phase 1: Iterative Development Testing

1. A development build was deployed.
2. Participants conducted structured play sessions.
3. Campaign testing logs documented usability and gameplay issues.
4. Regression testing verified stability after modifications.
5. Revisions were implemented before the next cycle.

This phase continued until a stable release candidate was achieved.

3.5.2 Phase 2: Final Structured Evaluation

1. Participants were briefed on the study purpose.
2. Consent was obtained.
3. Participants were given 20–30 minutes to interact with the finalized build.
4. Upon completion, participants completed:
 - MEEGA+KIDS instrument
 - Qualitative feedback questionnaire
5. Responses were collected via Google Forms.

3.6 Data Analysis

Quantitative data from the MEEGA+KIDS instrument were tabulated and analyzed using Google Sheets.

The following descriptive statistics were computed:

- Mean
- Standard deviation
- Frequency distribution
- Percentage scores

Results were interpreted to determine overall engagement and usability levels of the finalized build.

Data from campaign and regression testing logs were analyzed descriptively to identify:

- Reduction in bug frequency across iterations
- Improvement in gameplay stability
- Decrease in recurring usability issues

Qualitative responses were grouped into thematic categories to identify common patterns in player feedback.

No inferential statistical comparisons between builds were conducted, as iterative development phases involved varying participant groups.

3.7 Ethical Considerations

Participation in the study was voluntary. Participants were informed of the study's purpose and their right to withdraw at any time without penalty.

No personally identifiable information was included in the analysis. For participants under the age of 18, parental consent was secured prior to participation.

All collected data were used solely for academic research purposes.

CHAPTER 4

DESIGN AND IMPLEMENTATION

CHAPTER 5

GAME SYSTEM

5.1 Software Overview

Wiz's Journey: Mastering the States of Matter is developed using Unity 2022 LTS, a cross-platform game engine selected for its extensive capabilities in 2D game development, robust documentation, and the development team's prior expertise. The game is designed as a 2D platformer that integrates educational content with interactive gameplay mechanics, allowing students to progress through levels while encountering learning outcomes related to phase changes of matter.

The game is deployed across multiple platforms to ensure widespread accessibility and reach diverse student populations. The primary deployment target is Android mobile devices, distributed through Itch.io, a popular game distribution platform that allows easy access without requiring app store approval processes. A secondary web-based version is deployed via Unity Play, allowing students to play directly in web browsers on computers and tablets without installation requirements. This multi-platform approach reflects the varying technological infrastructure available to schools and students in the Philippines and aligns with the goal of making the educational game accessible regardless of device availability.

The development timeline began in December 2025 at the conclusion of the first trimester and extended through March 2026, spanning the academic term break and into the second trimester. This timeline was established to accommodate the academic calendar constraints and allowed the development team to complete the game while managing other coursework obligations. By the time development concluded in March 2026, the target Grade 6 students had already completed their first quarter instruction on phase changes under the MATATAG curriculum, positioning the game as a supplementary learning reinforcement tool rather than a primary instructional resource.

The game architecture incorporates both lower-order and higher-order mechanics aligned with the selected learning outcomes. Lower-order mechanics focus on foundational understanding of the four phase changes—melting, freezing, evaporation, and condensation—using visual representations and simple interactions. Higher-order mechanics build upon this foundation to address more complex concepts such as the relationship between heat transfer and state changes, the particulate nature of matter during phase transitions, and critical thinking about real-world applications of phase changes. The system integrates a particle lattice visualization system to support students' conceptualization of microscopic-level processes underlying

macroscopic observations, addressing a common misconception identified in science education research.

CHAPTER 6
RECOMMENDATIONS